Autonomous Karting Series Official Rulebook

Minor Changes

Changes in this section are not yet in the official rulebook but overwrite existing rules. The latest revision of a rule change will take precedence over past rule changes.

- Changes
  - Jan 25, 2026

Ruleset Table of Contents

- Administrative
  - Who Is Allowed To Compete
  - Required Roles
  - Documentation Requirements From Teams
  - Jurisdiction Of Rules
  - Disclaimer
- Electrical
  - Drivetrain
  - Batteries
  - General Wiring
  - Computers and Sensors
  - Design Documentation
- Software
- Mechanical
  - Frame / Chassis
  - Tires
  - Brakes
  - Weight
  - Mounting

    - Materials
    - BOM (Bill of Materials)
    - Resources
        - Go Kart Suppliers
        - Tires

- Safety
    - General Safety
    - Required Safety Components
    - Remote Control
    - Enforcement
    - Resources
        - Bodywork
        - Long Range Control (Including Remote EStop)

- Race Procedure
    - Spirit of the Competition
    - Event Structure
    - Race Track Definition
    - Allowed Data During Race
    - Multi-Kart Exposition Runs
    - Awards

- Appendix

Last Updated: Feb 3rd, 2026

Overview

This repository contains the development pages and official document for the Autonomous Karting Series Competition Rules. The individual sections of the rules have a folder with a markdown file for development and discussion purposes. The official document will be a protected latex file that requires approval to edit and modify.

Development Rulebook can be seen here

Changes

Disclaimer

This section covers changes to the officially released rules for one or more of the following purposes:

- Unanymous requests for changes from Team Leads that highlight issues that would prevent teams from competing.
- Ammending rules to clarify the spirit of the rule. These changes should not impact any team's ability to compete.

Changes in this section are not yet in the official rulebook but overwrite existing rules. The latest revision of a rule change will take precedence over past rule changes.

Update 2026_01_25

Reason For Change

Categorie(s):

- Ammending rules to clarify the spirit of the rule. These changes should not impact any team's ability to compete.

Purpose:

- Components that are not intended to be removed frequently from the chassis should not use the weakest form of threadlocker to satisfy this rule. Threadlocker on these semi-permanently mounted components will help prevent loosening from vibration during operation.

Rule(s) Changed

Mechanical > Mounting

```
-  1. Any component mounted directly to the chassis not mentioned already must be
secured with a threadlocker.
+  1. Any component mounted directly to the chassis not mentioned already must be
secured with a Medium Strength thread-locker.
      * Components that are needed to be removed and remounted frequently may be
secured with a Positively Locking Nut.
      * Electrical wires must be secured with cable ties or similar items.
```

Mechanical > Weatherproofing

```
   1. All components mounted to the go-kart chassis that are not rated for
exterior use must be contained within an IP65 Rated Container.
        Containers must use IP65 Rated grommets and fittings for cables and
connectors.
        Containers must have ventilation.
   2. Examples of components that must be in weather-rated containers:
        Motor Controller (if not rated for external use)
        Batteries
        Computers
        Microcontrollers
        Breakout/Distribution Boards

+  3. External components that are rated for IP64 (or are generally seen as
"weatherproof" without an IP certification) are also allowed.
+  4. Teams who have complications meeting Weatherproofing Requirements may use
lower IP64 Ratings, however, there is still the expectations that barring severe
weather all go-karts are able to operate.
```

Electrical > Batteries

```
-  7. Auxiliary Battery System Cannot exceed 50V nominal.
+  7. Auxiliary Battery System Cannot exceed Drivetrain (Primary) Battery System.
+    * The intent of the Auxiliary Battery System is to power auxiliary devices
such as sensors, computers, and low-level control devices.
```

Safety > Required Safety Components

```
    Urgent Stop
        All teams must prove that they have a switch to trigger an Urgent Stop
that has the go-kart come to a safe stop.
-           Urgent Stop must trigger if a controller leaves range or disconnects.
        Urgent Stop must be commandable through software
        Methods of stopping:
            Automatic stopping on Software Command or Physical Controller trigger.
            Switching to RC control to stop.
    Remote Estop must be tied to at minimum a physical controller.
        Off the shelf Emergency Stop systems are allowed as long as they do not
interfere with the track and has a physical trigger.
        Remote Estop should initiate an Urgent Stop AND THEN Disconnect power to
the Drivetrain.
+           The Go-kart should be able to re-power and be able to get back to
driving without physical intervention.
+       Remote Estop must trigger if a controller leaves range or disconnects.
```

Safety > Required General Safety

```
+      3. Teams are not allowed to work on the Go-Kart while the Go-Kart is in the
following states:
+           * Go-Kart has energized motors. An energized system is a system that
has components powered on.
+           * Go-Kart is in Autonomous Mode.
```

Administrative

- Who Is Allowed To Compete
- Required Roles
- Documentation Requirements From Teams
- Jurisdiction Of Rules
- Disclaimer

Who Is Allowed To Compete

1. Teams are required to be affiliated with an Academic Institution.
    - If the Academic Institution does not allow public affiliations then this is waived as long as the team members are comprised of students from the aformentioned Academic Institution.
2. A team can consist of members from more than one Academic Institution, however any one person cannot be on multiple teams.
3. One Academic Institution can host multiple teams.
4. Organizations are allowed to sponsor and/or mentor a team but are not allowed to contribute significantly to the go-kart of a team via mentors.
    - Judgment of “significant contribution” is at the discretion of the competition organizers.

Required Roles

1. All teams must establish at least one member as their Autonomous Safety Officer.
    - The Autonomous Safety Officer will ensure that their team’s kart is operating in a safe manner on and off track.
2. All teams must establish at least one Designated Vehicle Operator.
    - A Designated Vehicle Operator is a separate person from a team’s Autonomous Safety Officer.
    - A Designated Vehicle Operator is responsible for controlling the team’s kart as the kart is being operated remotely.

Documentation Requirements From Teams

1. All participants must sign and submit a liability waiver, photo release, and any event specific documents.
2. All teams must submit the required design documentation during the race check-in.
3. Some events may require certain components and features to be demonstrated in person.
    1. Teams must be able to demonstrate all components and features in person if requested.
    2. Teams will be notified of specific demonstrations before the event.

Jurisdiction Of Rules

1. All announcements and comments posted on the official website of the competition are to be considered part of the official ruleset

Disclaimer

These rules are open to interpretation where specifics are not given. This means that, for example, a material that is stated to be banned in one section is banned in all sections except where explicitly allowed. Other sections may have exceptions that explicitly allow or disallow under specific circumstances for those sections.

Mechanical Rules

- Frame / Chassis
- Tires
- Brakes
- Weight
- Mounting
- Weatherproofing
- Materials
- BOM (Bill of Materials)
- Resources
  - Go Kart Suppliers
  - Tires

Frame / Chassis

1. Any Commercial Full Size “Sprint Kart” chassis may be used
   - Custom Chassis may not be built from scratch.
   - For new teams, the Blue Shock Race X3 E-Kart is the new suggested platform to build on.
   - The BSR X3 can be bought in the following configurations from GetShockt:
     - Full Kit including Power Unit (15KW Motor + Controller) and Battery (96V Li-Ion)
       - With Front Brakes - $12,500 + freight
       - Rear Brakes Only - $13,300 + freight
     - Chassis Only
       - With Front Brakes - $5,500 + freight
       - Rear Brakes Only - $4,700 + freight
2. All Chassis components must be documented on a BOM declaring the part number and supplier.
3. All Chassis must include mounting points for Front and Rear bumpers.
4. Mounting Components may be welded to the chassis of the go-kart.
   - The chassis integrity must not be compromised by the welding process.
   - The process of welding to the chassis must act to only add structural parts and must not involve the removal of any section.

Tires

1. Go-Karts must use the following tire sizes:
    - Front Tires - 10 x 4.50-5
    - Rear Tires - 11 x 7.10-5
2. Go-Karts must use the following compounds of slick tires:
    - Hoosier R60B
    - Vega XH3 Green (Hard Compound)
    - Vega XM3 White (Medium Compound)
    - *Other tires may be allowed as long as they are of similar durometer to the above compounds. Please contact competition organizers for clarification on allowable tire compounds.*
3. Go-Karts must have the Front Wheels or Wheel Hubs mounted to the spindle with a **Castle Nut and Cotter Pin** or with a **Locking Nut and Safety Wire**. (*Refer to the wiki for examples.*)
    - If teams choose to use Wheel Hubs then the tires must be secured to the Wheel Hub with a **Locking Nut**. Safety Wires are not required.

Brakes

1. Go-Karts must have a minimum of 1 mechanical braking mechanism.
    - Motor-Based braking only in lieu of a mechanical brake is prohibited.
2. Go-Karts must at minimum have brakes on both rear tires.
    - A rear axle with 1 brake assembly attached will count for both rear tires.
3. Teams are encouraged to have brakes on all 4 wheels of the kart.
4. All hydraulic brake lines must be braided lines with the appropriate fittings to prevent leaks.

Weight

1. There is no maximum weight.
2. There is a minimum weight of 275 lbs with all components attached to the kart.
    - It is advised that teams use corner scales to balance their Go-Kart. Common Go-Kart weight distribution is provided below they are not a requirement but a suggestion for a starting distribution.
        - Front to Back -> 43% to 57%
        - Left to Right -> 50% to 50%
        - For further information please see this article *not affiliated*.

- Suggested go-kart weight is above 300 lbs.

Mounting

1. Any component mounted directly to the chassis not mentioned already must be secured with a threadlocker.
    1. Components that are needed to be removed and remounted frequently may be secured with a **Positively Locking Nut**.
    2. Electrical wires must be secured with cable ties or similar items.
2. **Internally Mounted Components** must be securely mounted such that the components do not become detached or loose inside their container.
3. The above mounting rules are enforcable through visual inspection with physical inspections if there is concern.

Weatherproofing

The purpose of requiring weatherproof containers is to prevent interruption due to light weather fluctuations. There is no intent to run in Heavy/Severe weather. IP65 is rated to withstand dust intrusion and water intrusion up to a direct stream of water.

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1. All components mounted to the go-kart chassis that are not rated for exterior use must be contained within an **IP65 Rated** Container.
    - Containers must use **IP65 Rated** grommets and fittings for cables and connectors.
    - Containers must have ventilation.
2. Examples of components that must be in weather-rated containers:
    - Motor Controller *(if not rated for external use)*
    - Batteries
    - Computers
    - Microcontrollers
    - Breakout/Distribution Boards

Materials

1. Any non-Internally Mounted Component must not be subject to the following:
    - Acrylic

    - Any **Brittle Material**
        - Most PLA and ABS 3D prints are classified as brittle.
2. Non-Brittle materials contain but are not limited to:
    - Polycarbonate
    - Delrin
    - PETG
    - TPU

BOM (Bill of Materials)

1. Teams must provide a Bill of Materials for all mechanical components **NOT** mounted internally.
    - Each mechanical assembly must be represented by an individual section in the BOM.
2. A team is able to estimate the cost of a component if it is unsure of the total value of the component.
    - This must be stated if the cost is estimated.

Resources

The following resources are not affiliated with Autonomous Karting Series Inc. but are included to aid teams in finding and sourcing components.

Go Kart Suppliers

- Get Shockt - Preferred Distributor for North America
- Blue Shock Race X3 Racing E-Kart
- Comet Kart Sales
- Coyote Karts
- PMC Kart
- Top Kart USA Electric

Tires

- Hoosier R60B - Comet Kart Sales
- Vega XH3 Green (Hard Compound)
- Vega XM3 White (Medium Compound)

- Tire Removal/Mounting Tool - Comet Kart Sales
- Bead Breaking Tool - Comet Kart Sales

Electrical Rules

- Drivetrain
- Batteries
- General Wiring
- Computers and Sensors
- Design Documentation

Drivetrain

1. The **Drivetrain** of the kart must be entirely electric.
2. Teams are allowed to use Brushed and Brushless motors.
3. Teams are not limited to the number of motors driving the rear tires.
4. Teams are not permitted to use motors to drive the front tires.
5. The **Total Drivetrain Output** of the motor(s) driving the rear tires must not exceed 15KW.
    - Teams are permitted to use higher output motors as long as it is configured not to exceed the above value.
    - Total Drivetrain Output must be measured before motor controllers.
    - Teams are encouraged to seek out ways to record their Total Drivetrain Output in a manner that the data can be monitored without external systems.
6. The **Wiring Harness** for the Drivetrain must include a fuse for the maximum expected current draw for a team's configured Total Drivetrain Output.
    - *Example: If a motor allows 20KW then the fuse must be able to support the full 20KW regardless of the team's intended power output.*

Batteries

1. Teams are allowed up to two battery systems which are listed below.
    - **Drivetrain Battery System**
    - **Auxiliary Battery System**
2. Each battery system must have an accessible and clearly labeled Kill Switch.
3. Teams are allowed to utilize the following battery chemistries:
    - Lithium Polymer (LiPo)
        - Lithium Cobalt Oxide (LCO)
        - Lithium Manganese Oxide (LMO)

        - Lithium Iron Phosphate (LFP)
        - Lithium Nickel Manganese Cobalt Oxide (NMC)
        - Lithium Nickel Cobalt Aluminum Oxide (NCA)
        - Lithium Titanate (LTO)
    - Lithium Ion Cells (18650, 21700, etc.)
    - Lead-Acid
4. All batteries must satisfy the following criteria:
    1. Clearly labelled and secured in place.
    2. Safely constructed.
        - Batteries must be enclosed within a proper container that allows cell expansion.
            - Shrink-Wrap is not a suitable enclosure.
        - *See mechanical requirements for weatherproofing.*
    3. Sufficiently cooled.
        - Batteries enclosures must have ventilation.
    4. Cannot have any exposed connectors or contacts.
    5. Contains Short Circuit protections to disconnect power in the event of a short circuit.
5. Battery enclosures should contain an internal fire supression system.
    - *This rule is not yet enforceable, however, this is the intended direction.*
    - In Battery Extinguisher
6. **Drivetrain Battery System** Cannot exceed 100V nominal.
7. **Auxiliary Battery System** Cannot exceed 50V nominal.
8. Batteries must use Anderson SB® 175 connectors as the external connectors.
    - Alternative connectors may be allowed on special request from teams.
    - The alternate connectors must at minimum meet the Anderson Specifications.

General Wiring

1. All **Wiring Harnesses** must be contained without any wires hanging out.
    - Wires are not contained when they are able to easily get caught on something or loosely laying on the kart.
    - Wires are contained when they are tied down and/or enclosed in a wire loom that is tied down.
2. Internal wiring must be routed and internally mounted such that the wires do not get tangled.
    - *Example: All internal wiring must be “best-effort” neat and orderly.*
3. All wires must be rated appropriately for the power being drawn through them.
4. There should be no exposed contacts or connectors.

Computers and Sensors

1. There is no regulation on sensors
2. There is no regulation on computers
3. Computation must be entirely contained to the kart
    1. Karts are not allowed to send data to an outside computer for computation used while running.

Design Documentation

1. Teams are required to provide a design document for the electrical systems.
2. Electrical Design Documentation must include at minimum the following:
    1. A description of every **Wiring Harness**.
    2. A description of every battery system.

Race Procedure Rules

- Spirit of the Competition
- Event Structure
- Race Track Definition
- Allowed Data During Race
- Multi-Kart Exposition Runs
- Awards

Spirit of the Competition

The goal of the Autonomous Karting Series is to create an accessible and exciting autonomous go kart race. The race presents a realistic scenario in which teams need to build go-karts that will race and compete on a track autonomously. It is encouraged that teams will work to create innovative and performant hardware, electrical, and software solutions.

Event Structure

1. There will be multiple Official Practice Sessions throughout a Race Event and a Official Race Session.
    - Official Practice sessions will follow competition safety rules as if the practice session was a Race Session.
    - Official Race Session will be 1 kart running on the track at a time.
    - Race Events may have Unofficial Practice Sessions.
        - During Unofficial Practice Sessions, **Secondary Vehicles** may be used for data collection for development purposes only.
        - **Secondary Vehicles** are not allowed on the track at any other time during the Race Event.
2. Team rankings will be by total time taken to complete the 5 laps.
3. The team order will be decided in order of fastest laps ran during official practice sessions.

Race Track Definition

1. The track is defined by the paved surface of the track
    - Individual races may have extra/special features to define the track.
2. Track width must be at minimum 15 feet.
3. The start/finish line will be identified on a per track basis.

Allowed Data During Race

1. ALL data collected during the Official Practice Sessions is allowed during the Race Session.
2. ALL data must be collected using the go-kart competing.
    1. Data collected using a secondary vehicle is not allowed.
3. A non-encompassing list of allowed data:
    - GPS waypoints
        - Track centerline
        - Track edges
    - Track images
    - Track pointcloud
4. Teams are encouraged to not focus entirely on a predefined run of the track but also take into account track conditions and adapt.
    - This will lead into full multi-kart races.

Multi-Kart Exposition Runs

1. During the event, there may be 1 or more exposition runs where 2 will be incentivized to showcase multi-kart racing capabilities under fully autonomous conditions.

Awards

1. Fastest Time Awards
    - The top 3 teams will receive placement awards in order of Race Total Time.
2. Engineering Excellence Award
    - The awarded team will be selected from judges based on design documents and implementation of components on the kart.

Safety Rules

- General Safety
- Fire Safety
- Required Safety Components
- Remote Control
- Enforcement
- Resources
    - Bodywork
    - Long Range Control (Including Remote EStop)

General Safety

1. Teams are not permitted on track while Go-Karts are running.
    - If a team needs to retrieve a Go-Kart then the team must wait for all Go-Karts to stop on track and the team has been given approval from race officials.
2. Teams are expected to be able to operate their go-karts completely remotely to reduce risk of on-track interventions.
    - Expected Remote Operations Include:
        - Resetting/Restarting Systems.
    - *Read As: Teams are encouraged to reduce the need for manual resetting of anything on the go-karts*

Fire Safety

1. All teams must have at minimum 2 fire extinguishers.
    - Fire Extinguishers must be rated for the type of batteries a team is using.
    - If a team is using a combination of battery compositions then the team must have extinguishers for each type of battery.
2. Examples of Fire Supression allowed:
    1. Lead-Acid Batteries:
        - Cat Litter
        - Sand
        - Class ABC Fire Extinguisher
    2. Lithium Batteries:
        - Class D Fire Extinguisher

- Specialized Lithium Extinguisher - E-FireX *not affiliated*

Required Safety Components

1. Urgent Stop
    - All teams must prove that they have a switch to trigger an Urgent Stop that has the go-kart come to a safe stop.
        - Urgent Stop must trigger if a controller leaves range or disconnects.
    - Urgent Stop must be commandable through software
    - Methods of stopping:
        - Automatic stopping on Software Command or Physical Controller trigger.
        - Switching to RC control to stop.
2. **Remote Estop** must be tied to at minimum a physical controller.
    1. Off the shelf Emergency Stop systems are allowed as long as they do not interfere with the track and has a physical trigger.
    2. **Remote Estop** should initiate an Urgent Stop AND Disconnect power to the **Drivetrain**.
3. **Physical Estop** that is clearly labelled and cuts all power to the kart.
    - Physical switch must exist on go-kart to disconnect ALL power on the go-kart
        - Minimum of 1 switch, if there are multiple switches then any 1 switch must disconnect all power.
        - Must be a Push-Button style Emergency Stop Switch.
        - Can be remote controlled –not required
    - Team must explain all kill-switches in inspection
4. Moving Commponents must have covers to prevent risk of injury or damage.
    - Chains must have chain covers.
5. There must be no exposed sharp edges on the kart.
6. Front and Rear bumpers are required.
    - Bumpers cannot be custom made and must be purchased.

Remote Control

1. Complete Remote Control is required.
    - Teams must be able to control the go-kart from their pit area regardless of go-kart position on the track.
    - Remote Control encompasses the following actions:
        - Triggering Safety Systems
        - RC Driving

    - Switchting between RC and Autonomous modes.
2. The **Remote Estop** controller is required to reach at minimum 200 meters line of sight to the kart.

Enforcement

1. Before a kart is allowed to run on the track it must undergo safety inspections and comply with all rules set forth in this document.
    1. Team Safety Officers must be present for that team's inspection.
2. Inspections may consist of the following:
    1. Visual inspections.
    2. Physical inspections.
    3. Document inspections.

Resources

The following resources are not affiliated with Autonomous Karting Series Inc. but are included to aid teams in finding and sourcing components.

Bodywork

- Front Noses - Comet Kart
- Rear Bumpers - Comet Kart

Long Range Control (Including Remote EStop)

- Crossfire Betaflight Article - Describes a solution for how to achieve long distance communication utilizing a 900MHz frequency.
- FrSky R9M Solution - Describes a solution utilizing used LoRa 900MHz frequency to achieve long distance.
- Kar-Tech Wireless Estop System - Off the shelf solution for Remote Estop.
- Radiomaster Pocket Radio Controller - ELRS - RadioMaster RC Controller with long-range capabilities
- Radiomaster RP4TD ExpressLRS - RadioMaster RC Receiver with long-range capabilities

Software & Network Rules

- Network
- RCS Black Box

Network

1. Teams are not allowed to use their own personal network devices to communicate with the go-kart from the pit.
2. Each team will be provided an ethernet drop that will be able to connect to the go-kart on the track.
3. Each team is expected to have a Ubiquiti Bullet AC for communication on the track.
    - Non weatherproof version - *use at your own risk*
    - *More information will be provided in how to setup the bullet network in the wiki.*

RCS Black Box

The Race Control System Black Box is a standalone device (with its own internal battery) that will provided to teams at the competition for use of managing go-kart and track safety. All software and designs will be made public. The Black Box network will be completely managed by AKS officials, is 100% visible to teams, and will only handle information described below.

---

1. Teams are expected to be able to interface with RCS Black Box via CANBus or USB.
2. Teams are expected to send the following information **TO** the RCS Black Box
    - Motor Output Power to the Drivetrain
    - Health Status - *status options to be provided*
    - Control Mode - **Autonomous** or **RC**
    - Speed - **m/s**
    - Steering - *may not be required*
3. Teams will be provided the following information **FROM** the RCS Black Box.
    - Track State - *states options to be provided*
    - Allowed Control Mode - **Autonomous** or **RC**
    - GPS RTCM
4. Teams will be provided a software library to integrate into their existing autonomy stack.

- The library will be available as a C++/Python library or as a ROS2 package.

Appendix

Definable Terms

- BOM - Bill of Materials
- Internally Mounted Components - a component or collection of components enclosed completely inside something mounted directly to the chassis.
- Brittle Materials - materials that are prone to shattering and/or creating stiff sharp edges upon breaking.
- Drivetrain - the motor(s) and motor controller(s) that drive the kart.
- Total Drivetrain Output - the total power output in kilowatts of all motors on the drivetrain.
- Wiring Harness - a collection of wires and connectors for a specific component or grouping of components.
- Remote E-stop - a remotely controlled emergency stop system that can be triggered if the kart becomes unstable.
- Physical E-stop - a physical button or switch that can be activated if the kart becomes unstable.
- Autonomous Driving - when the kart is being controlled only by the pre-programmed logic contained on the kart.
- RC Driving - when the kart is being controlled by a remote controller exterior to the kart.
- Pre-Mapping - utilizing data collected prior to a race start with the intent to use it to compete in the race.
- Secondary Vehicles - any vehicle that is not the competing go-kart and is controlled remotely.